1.4.2 Distributive Property of Multiplication over Addition

We have already studied that multiplication distributes over addition on whole numbers. Let us check the property for integers.

Take for example $(-2)\times(4+5)=\big[(-2)\times4\big]+\big[(-2)\times5\big]$

LHS	RHS
$= (-2)\times(4+5)$	$= \big[(-2)\times4\big]+\big[(-2)\times5\big]$
$= (-2)\times9$	$= (-8)+(-10)$
$= (-18)$	$= -8-10$
$= -18$	$= -18$

From the above, we can observe that $(-2)\times(4+5)=\big[(-2)\times4\big]+\big[(-2)\times5\big]$

Hence, "distributive property of multiplication over addition" is true for integers.

Therefore, for any three integers a, b, c ; $a\times(b+c)=(a\times b)+(a\times c)$

Try these

1. Find the values of the following and check for equality:
   (i) (–6) × (4+(–5)) and ((–6) ×4) + ((–6)×(–5))
   (ii) (–3)×[2+(–8)] and [(–3)×2]+[(–3)×8]
2. Prove the following:
   (i) (–5) × [(–76)+8] = [(–5) ×(–76)] + [(–5)×8]
   (ii) 42 × [7+(–3)] = (42 ×7) + [42×(–3)]
   (iii) (–3) × [(–4)+(–5)] = ((–3) × (–4)) + [(–3)×(–5)]
   (iv) 103 × 25 = (100+3) × 25 = (100×25) +(3×25)

Example 1.20

Prove that (–7) × (+8) is an integer and mention the property.

Solution

(–7) × (+8) = (–56)

Hence, –56 is an integer.

Therefore, (–7) × (+8) is closed under multiplicaton.

Example 1.21

Are (–42) × (–7) and (–7) × (–42) equal? Mention the property.

Solution

Consider, (–42) × (–7),

(–42) × (–7) = +294

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